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DEPARTMENT OF GENETICS AND BIOENGINEERING
BIOE 401 - Bioengineering Design

Glucosamine Production Analysis

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Equipment sizing:

Table 1. Equipment specifications

Equipment	Specifications
Fermenters (2)	The parts in contact with the material are SS316L, and the others are SS304. Internal roughness less than 0.4µm, polished Different according to different capacity, can also be customized Floor-type One fermentor vessel with one control system (Siemens PLC and HMI) Capacity: 7500 L Temperature: 37°C pH: 6.8 - 7.0
Tubular Centrifuge (2)	• max. Speed: 30000 rpm • max. RCF: 65395 xg • max. Volume: 6 x 250 ml • Speed range: 200 - 30000 rpm • Temperature range: -20°C to 40°C • Dimensions (WxHxD): 71,5 cm x 42 cm x 51 cm • Weight: 91 kg
Homogenizer	Capacity: 750 L for cells 6750 L broth
	Pressure: 37 MPa
Ion exchange column	Capacity: 8000 L
	Temperature: 100 °C
	pH: acidic (HCl)
Basket Centrifuge	Capacity: 750 L for cells 6750 L broth
	Rotational Speed: 3,000 to 6,000 RPM
	Temperature: 4 °C
Vacuum Dryer	Capacity: 700 L
	Temperature: 40 - 50 °C
	Pressure: 20 - 50 mbar

CAPITAL COST:

Estimate using Bridgewater's Method:

$$Q < 60,000: \quad C = 280,000 N \left(\frac{Q}{s} \right)^{0.3}$$

- $Q = 57.5$ metric tons per year
- $N = 7$
- $S = 0.94$
- $C = 280,000 (16)(57.5/0.94)^{0.3} = \mathbf{15,390,049.3 \text{ \$/year}}$

MANUFACTURING COST

Manufacturing cost equals the sum of Direct Operating Costs, Fixed Costs, and Plant Overhead Costs.

$$\text{Manufacturing cost} = 6,383,376.47 + 4,094,473.75 + 76,016.25$$

$$\text{Manufacturing cost} = \mathbf{10,553,866.47 \text{ \$/year.}}$$

Each of the costs was calculated as follows:

1) Direct Operating Costs:

a) Raw material and consumables cost:

The cost of the raw materials and consumables (for both upstream and downstream manufacturing) was gathered from various sources (raw material and consumables prices websites: Thermo Fisher Scientific, Sigma Aldrich, Alibaba, Research Products International, export Hub, and FLINN Scientific). The cost of the antifoam agent was estimated to be \$20 per kilogram. Tables 2 and 3 display the cost of raw materials and consumables used in the large-scale upstream and downstream manufacture of glucosamine.

The Total cost of raw materials is \$ 9,957.186 per batch, and The Total cost of consumables is \$34,362.15 per batch, the ion exchange resin was excluded from the total cost since it is reusable.

The number of batches per year is 69.77, thus The Total cost of Raw Materials Cost per Year is 694,712.87 \$/year, and the total cost of Consumables Cost per Year is 2,397,447.21 \$/year.

Table 2. Upstream process estimates costs

Raw Materials/Consumables	Quantity per batch (Kg)	Price (\$/kg)	Cost per batch (\$)
<u>Raw Materials:</u>			
Glucose (65% purity)	5,017.5	0.66	3,311.6
Potassium phosphate monobasic	65	4.3	279.5
Citric acid monohydrate	15	0.72	10.8
Calcium chloride	0.502	2.8	1.41
Magnesium sulfate heptahydrate	50.18	0.7	35.126
Lactose	37.5	3.5	131.25
Deionized Water	7,500	0.8	6000
E. coli strain K-12	1 strain	187.50	187.50
<u>Consumables:</u>			
Antifoam (DF204a)	5.02	20	100.4
Ammonium Hydroxide	750	0.33	247.5

Table 3. Downstream process estimates costs (Consumables)

Consumables	Quantity per batch (Kg)	Price (\$/Kg)	Cost per batch (\$)
Polyethylenimine	7.5	32.50	243.75
Hydrochloric Acid (2N)	2967	1	2,967
Ethanol (95%)	2465	13.7	33,770.5
Dow 650C resin	1567	39.73	62,256.91

The total cost of raw materials and consumables is 3,092,159.87 \$/year.

b) Cost of utilities:

Utilities are typically calculated as a percentage of equipment-dependent costs which is equal to 2,533,875 \$ from Table 5.

Assuming 5% of equipment-dependent costs:

$$\text{Utilities cost} = 0.05 \times 2,533,875 \$ = 126,693.75 \$/\text{year}$$

c) Wastes treatment:

Waste treatment costs are typically estimated as 5% of raw material costs.

$$\text{Waste disposal cost} = 0.05 \times 694,712.87 \$/\text{year} = 34,735.65 \$/\text{year}$$

d) Labor:

Labor costs are estimated based on a fixed percentage of the revenue. This percentage can vary by industry but typically ranges from 10% to 30%. Assuming 20% of the revenue the labor cost is 3,129,787.2 \$/year.

Total Direct Operation Cost is equal to the sum of Raw material, consumables, utilities, Waste disposal, and Labor costs, which is equal to **6,383,376.47 \$/year**

2) Fixed Costs:

a) Depreciation

Depreciation is 5 % of the direct cost of capital investment.

- Direct Cost: 7,511,755 \$ from Table 6.
- Depreciation = $0.05 \times 7,511,755 = 375,587.75 \$/\text{year}$

b) Taxes (5%) and Insurance

Taxes are assumed to be 5% of the total operating cost.

$$\text{Taxes} = 0.05 \times 6,834,980.47 = 341,749 \text{ \$/year}$$

Insurance is included in the indirect costs and not separately itemized

c) Rent/ Building cost: The total building cost is 3,377,137 \$ from Table 4.

Table 4 Building cost estimation

Space function	Unit cost (\$/m ²)	Cost (US\$)
Class 3000 facility	1000	1,133,500
Mechanical room	15	1,541.25
Laboratory	16	2,054,000
Office	16	188,096
Total	3,377,137	

The Total **Fixed Cost** is equal to **4,094,473.75 \$/year**

3) Plant Overhead Costs (Service Charges for Maintenance and Repair) :

Maintenance and repair are part of the 3% equipment-dependent costs.

Equipment-dependent costs: 2,533,875 \$ /year

$$\text{Maintenance and repair cost} = 0.03 \times 2,533,875 = \mathbf{76,016.25 \text{ \$/year}}$$

Estimate Capital Investment Cost

- 1) Estimation of Purchased Equipment Cost:** The purchase cost of major components for the glucosamine process is collected and summarized in Table 5 based on the equipment specification in Table 1.

The total equipment purchasing cost is **2,533,875 \$**

Table 5. Total equipment purchasing cost

Equipment	Purchase Cost (US \$)
7500 L fermenters (2)	600,000
6000–8000 L vessels (3 total)	1,059,375
Homogenizer	75,000
Centrifuge (2)	29,500
Ion exchange column	200,000
Basket Centrifuge	500,000
Vacuum Dryer	70,000

2) Estimating FCI and Working Capital Costs using the Factorial Method:

Multiply the total equipment purchase cost by fixed factors to estimate fixed capital investment (FCI) and Working Capital.

The total Fixed Capital Investment cost equals the sum of the direct, indirect, and start-up costs, equaling 8,511,755 \$.

The working capital cost is \$ 438,000. Costs calculated using the Factorial Method are summarized in Table 6.

The **Total capital investment**, the sum of the total fixed capital investment cost and working capital costs is equal to **14,336,611 \$**

Table 6. Total Capital Investment estimated cost for the glucosamine process

Type of cost	Cost (US \$)
<u>Direct cost:</u>	
Equipment	2,533,875
Installation	890,000
Process piping	312,000
Instrumentation	50,000
Insulation	60,400
Electrical	289,000
Buildings ¹	3,377,137
Yard improvement	289,000
Auxiliary facilities	890,000
	= 8,511,755
<u>Indirect cost:</u>	
Engineering	1,100,850
Construction	2,141,190
Contractor's fee	544,272
Contingency	1,022,544
	=4,808,856
Start-up Costs	578,000
Working capital	438,000

¹ Estimated in Table 4

Operating Costs:

Operating costs include:

- Raw materials cost
- Consumables cost
- Labor cost

- Utilities cost
- Waste disposal cost
- Depreciation

Operating Cost= Direct Operation cost + Depreciation + Maintenance and Repair Costs

Substituting values:

Total Operating Cost = 6,383,376.47 + 375,587.75 + 76,016.25

Operating Cost = 6,834,980.47 \$/year

Thus, the **Total Operating Cost** for the process is **6,834,980.47 \$/year**.

REVENUES:

The revenues for the glucosamine production are calculated based on the annual production capacity and the selling price of glucosamine:

- Annual Production Capacity: 57.5 tons/year = 52162.13 kg/year
- Selling price: \$300/kg

Revenues can be calculated as:

Revenues = Annual Production Rate × Selling price

Revenues = 52162.13 kg/year × 300 \$/kg = **15,648,639 \$/year**

Payback Period:

The payback period is determined using the formula:

Payback Period = Total Capital Investment / Net Annual Profit

- Total Capital Investment: \$ 14,336,611

- Gross Profit = Revenue - Manufacturing cost = 15,648,936 - 10,553,866.47 = 5,094,772.53 \$/year
- Taxes = Gross Profit $\times$ 0.2 = 5,094,772.53 \$/year $\times$ 0.2 = 1,018,954.51 \$/year
- Net Annual Profit = Gross Profit-Taxes= 5,094,772.53 -1,018,954.51= 4,075,818.02 \$/year

Substituting:

$$\text{Payback Period} = 14,336,611 / 4,075,818.02 \approx 3.52 \text{ years}$$

Cash Flow diagram:

The cash flow graphic shows the glucosamine manufacturing process's financial performance under three distinct operating conditions: higher fermentation titer, lower ethanol recovery, and the base case (ideal conditions). To show how important process factors affect profitability and payback time, each scenario is a sensitivity analysis.

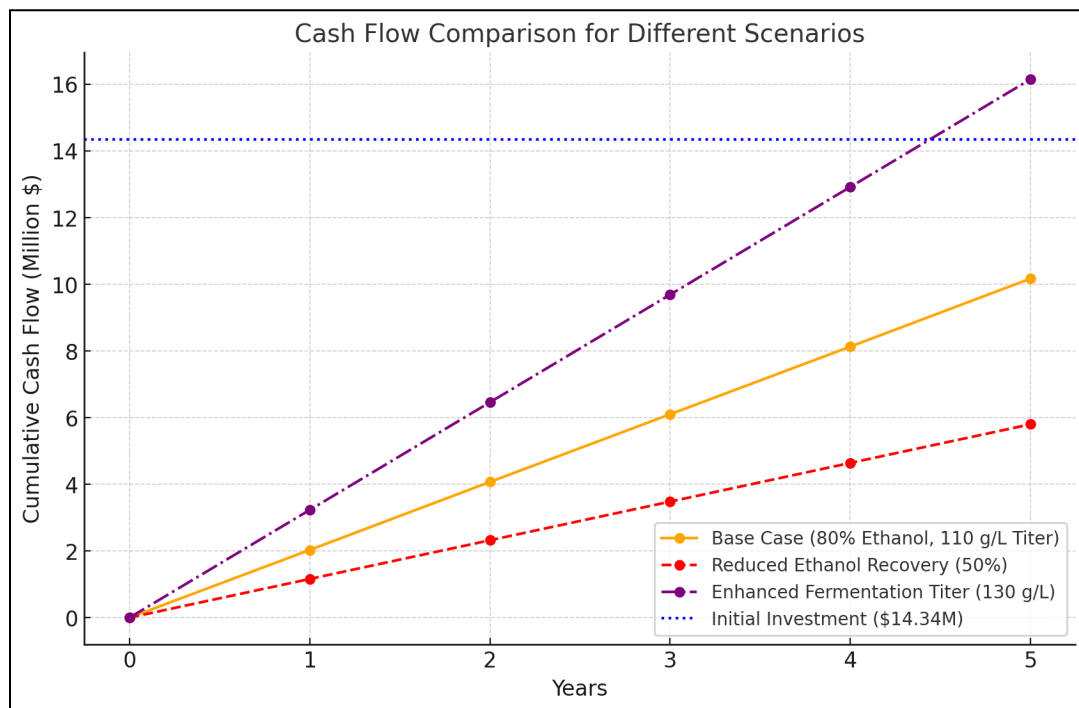


Figure 5.1: Cash Flow Graph

Based on the cash flow comparison graph, here is an analysis of the different scenarios:

- Base Case (80% Ethanol, 110 g/L Titer):

This is the baseline scenario, where the ethanol recovery rate is 80% and the fermentation titer is 110 g/L. The cumulative cash flow increases steadily over the 5 years, reaching around \$9 million by the end of the 5th year.

- Reduced Ethanol Recovery (50%):

In this scenario, the ethanol recovery rate is reduced to 50%, which increases production costs due to higher ethanol consumption and decreased recycling effectiveness. The cumulative cash flow increases at a slower rate compared to the base case, reaching around \$6.5 million by the end of the 5th year. The payback period is extended to around 5.6 years.

- Enhanced Fermentation Titer (130 g/L):

This scenario assumes an improved fermentation titer of 130 g/L, compared to the base case of 110 g/L. The higher production yield leads to increased profitability, as reflected in the steeper slope of the cumulative cash flow curve. The payback period is shortened to around 3.8 years, with the cumulative cash flow reaching approximately \$12.5 million by the end of the 5th year.

References:

- <https://patentimages.storage.googleapis.com/2d/df/eb/d2d0b0599dd10a/US20050148546A1.pdf>
- <https://www.davuniversity.org/images/files/study-material/PLANT%20DESIGN%20AND%20ECONOMICS%20FOR%20CHEMICAL%20ENGINEERS.pdf>
- <https://assessccus.globalco2initiative.org/wp-content/uploads/Factorial-Techniques.pdf>